13.7 Manufacturing Correlation

WORKSHEET

The question

STEP III 2010, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s3-q13

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Z_1 and Z_2 are independent. What is their covariance, and so their correlation?

STAGE 1 Two variables

Ask yourself Think of Z_1 and Z_2 as perpendicular unit vectors. Where is Y_2? What is the cosine of the angle between Y_1 and Y_2?

STAGE 2 The third variable

Ask yourself Write down the three conditions $\text{Var}(Y_3) = 1$, $\text{Cov}(Y_1, Y_3) = \rho_{13}$, $\text{Cov}(Y_2, Y_3) = \rho_{23}$ in terms of a, b, c.

STAGE 3 Any means and variances

Pause and try The formula for c needs the quantity under the square root to be non-negative. Try $\rho_{12} = \rho_{13} = \rho_{23} = -0.6$. What goes wrong, and why does it make sense that three variables cannot all be strongly negatively correlated with each other?

What's really going on

The Cholesky decomposition

In matrix form, the construction says $\mathbf{Y} = L\mathbf{Z}$ with a *lower triangular* matrix

$$L = \begin{pmatrix} 1 & 0 & 0 \\ \rho_{12} & \sqrt{1-\rho_{12}^2} & 0 \\ a & b & c \end{pmatrix},$$

and the correlation matrix of $\mathbf{Y}$ is $R = LL^{\mathsf{T}}$. Writing a symmetric matrix as LL^{T} is the *Cholesky decomposition*, found by the French military engineer André-Louis Cholesky around 1910 for solving the equations of geodetic surveys (published after his death in the First World War). It works whenever R is *positive definite*, and the computation is exactly what we did by hand: each row is solved in turn. It is the standard way to simulate correlated variables in finance, engineering and climate modelling.

Which correlations are possible?

The number under the square root for c must be non-negative. Multiplying out, the condition is

$$1 - \rho_{12}^2 - \rho_{13}^2 - \rho_{23}^2 + 2\rho_{12}\rho_{13}\rho_{23} \geq 0,$$

which is the condition that the determinant of the correlation matrix is non-negative. Geometrically: three unit vectors with pairwise angles θ_{ij} (where $\cos\theta_{ij} = \rho_{ij}$) must actually fit together in space. If $\rho_{12} = \rho_{13} = \rho_{23} = -0.6$, the angles are all about 127°, and three vectors at 127° to each other would need angles summing to more than 360°: impossible. In general a matrix can be a correlation matrix only if it is positive semi-definite, all its eigenvalues non-negative (Section 7.2). When correlations are estimated separately from incomplete data, this can fail, and statisticians must find the nearest valid correlation matrix, a problem solved by Nicholas Higham in 2002.

Galton, Pearson and correlation

Francis Galton introduced correlation in 1888, while studying how the heights of parents and children are related, and Karl Pearson gave the product moment formula used in this question in 1896. Galton also noticed regression to the mean (Section 13.1). Pearson founded the world's first university statistics department, at University College London, in 1911. The warning that "correlation does not imply causation" is almost as old as correlation itself.

Try this next

1. **Impossible correlations.** Show that three random variables cannot all have pairwise correlation ρ unless $\rho \geq -\frac{1}{2}$. What is the corresponding statement for n variables?

2. **Check the determinant.** Show that $1 - \rho_{13}^2 - b^2 \geq 0$ is equivalent to $\det R \geq 0$, where R is the 3×3 correlation matrix.

3. **Portfolio risk.** Two shares have returns with standard deviations $\sigma_1 = 0.2$ and $\sigma_2 = 0.3$ and correlation ρ. Find the standard deviation of a portfolio half in each, for $\rho = -1$, 0 and 1. What does diversification achieve?

4. **Recover the ingredients.** Given Y_1, Y_2 as in the question, express Z_2 in terms of Y_1 and Y_2. (This is the Gram–Schmidt step.)

Further reading

- **Gilbert Strang,** ***Introduction to Linear Algebra*** (Wellesley-Cambridge), on positive definite matrices and the Cholesky factorisation.
- **Paul Glasserman,** ***Monte Carlo Methods in Financial Engineering*** (Springer). Simulating correlated variables in practice; advanced but the early chapters are readable.
- **Theodore Porter,** ***Karl Pearson: The Scientific Life in a Statistical Age*** (Princeton University Press).

The full guided solution is in Section 13.7:
https://beyondstep.keithmcnulty.org/ch13/correlation.html